

AI/ML Quick Notes

1. Machine Learning

Machine learning allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

2. Supervised Learning

Supervised learning uses labeled training data. Common tasks include classification and regression.

3. Unsupervised Learning

Unsupervised learning works with unlabeled data and tries to discover hidden patterns or groups.

4. Classification

Classification predicts a category or class. For example, a model can classify an email as spam or not spam.

5. Regression

Regression predicts a continuous numerical value. For example, a model can predict house prices.

6. Overfitting

Overfitting occurs when a model learns the training data too closely, including noise, and performs poorly on new data.

7. Underfitting

Underfitting occurs when a model is too simple to learn important patterns in the data.

8. Model Evaluation

Model evaluation measures how well a machine learning model performs. Common metrics include accuracy, precision, recall, and AUC.

9. Feature Engineering

Feature engineering involves creating or selecting useful features from raw data to improve machine learning performance.

10. Generative AI and Large Language Models

Generative AI can create content such as text and images. Large language models generate text based on patterns learned from large amounts of data.